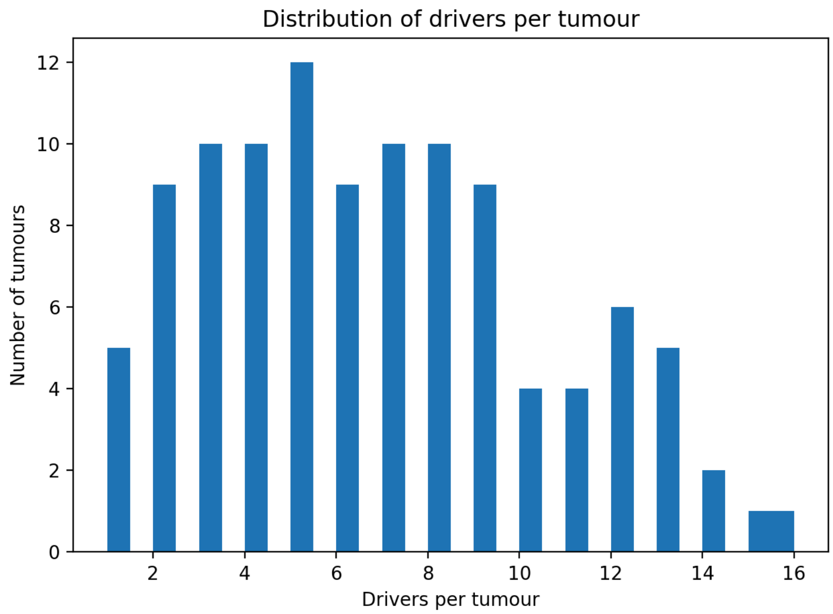
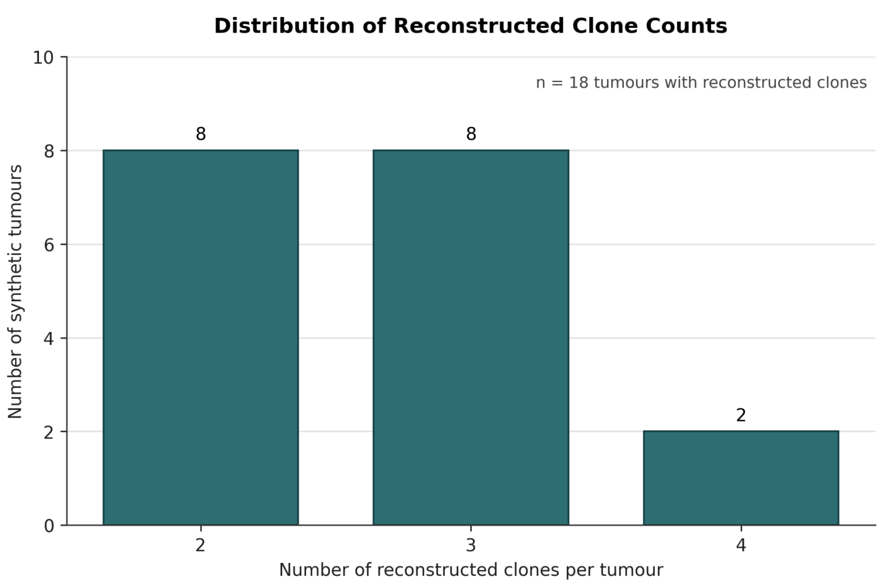
Supplementary Figures



Supplementary Figure 1. Empirical distribution of the number of driver genes per tumour derived from PCAWG/ICGC breast adenocarcinoma samples.



Supplementary Figure 2. Distribution of the number of reconstructed clones per synthetic tumour in the representative synthetic breast cancer cohort.

Supplementary Tables

Supplementary Table 1. Clinical characteristics of one representative synthetic breast cancer cohort generated by the pipeline.

Characteristic	Number of patients	Percentage	Notes
Total breast cancer patients	39	100	Patients with malignant neoplasm of breast diagnosis
Cancer Stage I	7	17.9	
Cancer Stage II	6	15.4	
Cancer Stage III	4	10.3	
Cancer Stage IV	3	7.7	20 patients in total had diagnostic staging recorded through the breast biopsy/TNM assessment pathway.
HER2-positive	6	15.4	Positive IHC or FISH greater than 2.2
ER-positive	16	41.0	Positive estrogen receptor IHC
PR-positive	12	30.8	Positive progesterone receptor IHC
Triple-negative phenotype	2	5.1	ER-negative, PR-negative, and HER2-negative
Latest treatment response: Improving	21	53.8	Latest recorded treatment response per patient
Latest treatment response: Worsening	4	10.3	Latest recorded treatment response per patient
Breast surgery	14	35.9	Lumpectomy or mastectomy
Chemotherapy	13	33.3	Chemotherapy

			medication or chemotherapy procedure exposure
Radiation therapy	18	46.2	Radiation therapy or brachytherapy procedure
Endocrine therapy	17	43.6	tamoxifen, aromatase inhibitor (letrozole, anastrozole, exemestane) or fulvestrant
HER2-targeted therapy	7	17.9	trastuzumab, lapatinib, or neratinib
CDK4/6 inhibitor therapy	11	28.2	palbociclib, ribociclib, or abemaciclib
Two or more tumour genomic profiling procedures	18	46.2	
Three tumour genomic profiling procedures	5	12.8	
Breast-cancer cause of death	6	15.4	

Supplementary Table 2. Genomic and clone-level characteristics of the same representative synthetic breast cancer cohort summarized clinically in Supplementary Table 1.

Domain	Characteristic	Result	Notes
Genomic testing	Patients with at least one tumour genomic profiling procedure	22 (56.4%)	Recorded in procedures.csv
Genomic testing	Patients with longitudinal clone reconstruction	18 (46.2%)	Had at least two tumour genomic profiling procedures. Clone-level MAF output was generated for these patients.
Genomic testing	Patients with exactly	13 (33.3%)	

	two tumour genomic profiling procedures		
Genomic testing	Patients with exactly three tumour genomic profiling procedures	5 (12.8%)	Maximum number of tumour genomic profiling procedures
Clone architecture	Reconstructed clones	48	Across the 18 patients who had at least a second tumour genomic profiling procedure
Clone architecture	Founding clones	18	One per reconstructed patient
Clone architecture	Non-founding clones	30	
Clone architecture	Patients with two clones	8	
Clone architecture	Patients with three clones	8	
Clone architecture	Patients with four clones	2	
Driver composition	Clone-level driver assignments	143	Across all clones
Driver composition	Distinct driver genes	34	
Driver composition	Driver genes per patient	Median 8 Range 4–14	
Driver composition	Driver genes per clone	Median 3 Range 1–5	
Driver composition	Most frequent clone-level driver assignments	PIK3CA (16), GATA3 (11), MAP3K1 (10), TP53 (8), ESR1 (8), CCND1 (7)	The counts indicate clones containing the gene
Driver composition	Other recurrent assignments	ERBB2/HER2, BRCA2, ZNF703 (6 each); CHEK2, NCOR1, MAP2K4 (5 each)	The counts indicate clones containing the gene
Enhanced observations	Genomics-related observation rows	1 170	In the enhanced observations output file

Enhanced observations	Structured clone-related annotations	234 DNA-change, 234 clonal-prevalence and 234 clonal-role rows	
MAF output	Clone-level MAF files	48	One MAF file per reconstructed clone
MAF output	Total MAF variant records	139 530	Drivers and passengers combined
MAF output	Driver variant records	143	Correspond to the 143 clone-gene assignments
MAF output	Passenger variant records	139 387	Derived from assigned ICGC breast cancer samples
MAF output	Total variants per clone	Range 466 – 27 379	Drivers and passengers combined
MAF output	Total variants per patient	Range 1 696 - 54 756	Sum across that patient's clones
Passenger burden	Passenger variants per clone	Median 1 607 Range 464 - 27 374	
Passenger burden	Passenger variants per patient	Median 5 008 Range 1 692 - 54 748	Among patients with MAF output

Supplementary Table 3. Drug-gene sensitivity and resistance associations used in the generator, with references.

Drug(s)	Association	Gene	Mutation type	References (each CIViC link points to a different molecular profile)
Epirubicin	Resistance	TP53	Disruptive/inactivating	(Aas *et al.* 1996)
Epirubicin	Resistance	CHEK2	Loss-of-Function	(Chrisanthar *et al.* 2011)
Doxorubicin	Resistance	TP53	Disruptive/inactivating	(Aas *et al.* 1996)
Doxorubicin	Resistance	CHEK2	Nonsense	(Chrisanthar *et al.* 2011)
Tamoxifen + Aromatase	Resistance	NF1	Loss of function	(Sokol *et al.* 2019)

Inhibitors (Letrozole, Anastrozole, Exemestane)				
Endocrine Therapies	Resistance	ESR1	Fusion	(Hartmaier *et al.* 2018)
Endocrine therapies	Resistance	RUNX1, CTCF, ARID1A	Gene disruptive	(Saatci, Huynh-Dam, and Sahin 2021)
Fulvestrant	Resistance	ARID1A	Loss of Function	(Xu *et al.* 2020)
Aromatase Inhibitors (Letrozole, Anastrozole, Exemestane)	Sensitivity	GATA3, MAP3K1, MAP2K4	Somatic coding mutations broadly	(Ellis *et al.* 2012)
Aromatase Inhibitors (Letrozole, Anastrozole, Exemestane)	Resistance	TP53	Functionally damaging TP53 alterations	(Ellis *et al.* 2012)
Fulvestrant	Resistance	ARID1A	Inactivating	(Xu *et al.* 2020)
Trastuzumab	Resistance	PTEN	Loss of Function	(Rimawi *et al.* 2018)
Trastuzumab	Resistance	PIK3CA	Activating mutation	(Rimawi *et al.* 2018)
Lapatinib	Resistance	PTEN	Loss of Function	(Rimawi *et al.* 2018)
Lapatinib	Resistance	PIK3CA	PIK3CA mutation / PI3K pathway activation	(Rimawi *et al.* 2018)
Palbociclib + Ribociclib + Abemaciclib	Resistance	RB1, FAT1	Loss of Function	(Li *et al.* 2018)
Palbociclib + Ribociclib + Abemaciclib	Resistance	PTEN, NF1	Pathway escape	(Huang *et al.* 2022)
Doxorubicin	Sensitivity	TP53	Missense Mutation and Protein Altering Variant	CIViC, CIViC
Fulvestrant	Sensitivity	PIK3CA	Various	CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC,

				CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC
Fulvestrant	Sensitivity	AKT1	Missense Mutation	CIViC
Fulvestrant	Sensitivity	ESR1	Missense Mutation	CIViC, CIViC, CIViC, CIViC
Trastuzumab	Sensitivity	PIK3CA	Missense Mutation	CIViC
Trastuzumab	Sensitivity	FCGR3A	Missense Mutation	CIViC, CIViC
Trastuzumab	Sensitivity	ERBB3	Missense Mutation	CIViC
Lapatinib	Sensitivity	PIK3CA	Missense Mutation	CIViC
Lapatinib	Sensitivity	ERBB3	Missense Mutation	CIViC
Neratinib	Sensitivity	ERBB2 (HER2)	Various	CIViC, CIViC,CIViC, CIViC
Palbociclib	Sensitivity	PIK3CA	Various	CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC
Palbociclib	Sensitivity	ESR1	Various	CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC
Tamoxifen	Sensitivity	ESR1	Various	CIViC, CIViC, CIViC, CIViC, CIViC
Letrozole	Sensitivity	ESR1	Missense Mutation	CIViC
Exemestane	Sensitivity	STK11	Missense Mutation	CIViC
Ribociclib	Sensitivity	PIK3CA	Transcript Variant, Gain Of Function Variant	CIViC
Abemaciclib	Sensitivity	ESR1	Not provided	CIViC
Tamoxifen	Resistance	TP53	Missense Mutation	CIViC
Trastuzumab	Resistance	PIK3CA	Missense Mutation	CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC, CIViC

Trastuzumab	Resistance	ERBB2 (HER2)	Not specified	CIViC, CIViC, CIViC
Lapatinib	Resistance	ERBB2	Various	CIViC, CIViC, CIViC
Lapatinib	Resistance	PIK3CA	Missense mutation	CIViC, CIViC, CIViC, CIViC, CIViC
Neratinib	Resistance	ERBB2	Missense mutation	CIViC
Palbociclib	Resistance	RB1	Frameshift Truncation	CIViC
Ribociclib	Resistance	RB1	Not specified	CIViC

Computational Requirements

Clinical-Genomic Cohort Generation

Benchmark setup

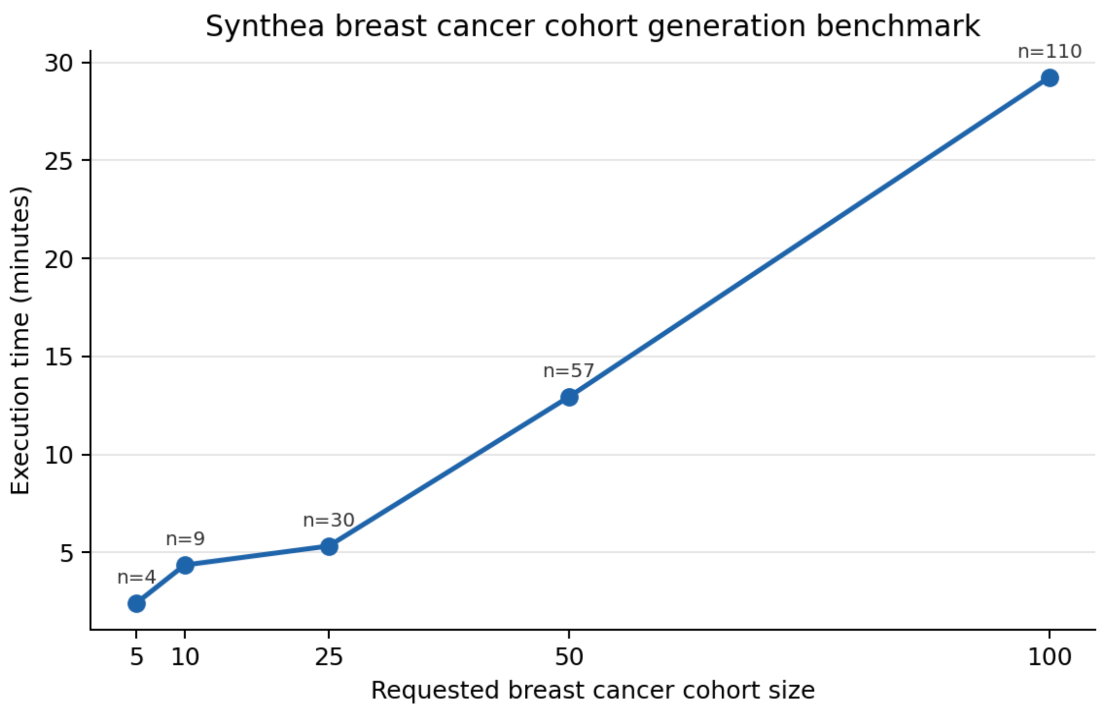
Computational requirements were evaluated for both Synthea cohort generation and the downstream breast-cancer genomics post-processing workflow. Benchmarks were run on a workstation with an Intel(R) Core(TM) Ultra 7 165U CPU, 14 logical CPUs, 15 GiB RAM, and OpenJDK 11.0.30. Runtime and peak resident memory were measured using GNU "/usr/bin/time -v", and disk usage was measured from the generated output directories after each run.

For Synthea generation, breast cancer cohorts were generated using the "src/main/resources/synthea.properties" configuration, the local module directory "src/main/resources/modules", and the breast cancer keep filter "src/main/resources/keep_modules/keep_breast_cancer.json", all of which can be found in the project repository. The same setup was used to generate the representative synthetic breast cancer cohort described in the Results section, summarized in Supplementary Tables 1 and 2, and provided in the project repository.

The age range was restricted to 45-90 years. This range concentrated candidate generation in middle-aged and older adults, in whom breast cancer is more frequent, thereby improving generation efficiency when applying the breast-cancer keep filter. To make runs comparable, the Synthea seed and clinician seed were both fixed to "12345", the reference date and end date were both fixed to "20260507" (May 7, 2026), and the Java heap limit was set to "-Xmx2g". The maximum number of candidate-generation attempts per retained patient was set to "1000". CSV export was enabled, while FHIR export was disabled. Cohort sizes of 5, 10, 25, 50, and 100 requested patients were benchmarked. The number of retained patient rows can differ from the requested population size when the keep-filter and maximum-attempt limit are applied, therefore, both requested size and generated patient rows are reported in the supplementary table below.

Synthea cohort generation performance

Synthea generation time increased with requested breast cancer cohort size (Supplementary Figure 3). Wall-clock runtime ranged from 2.42 minutes for a requested cohort of 5 patients to 29.24 minutes for a requested cohort of 100 patients. Peak resident memory ranged from 1.51 to 2.17 GB across these runs. CSV output storage ranged from 15.88 MB to 179.17 MB (Supplementary Table 4).



Supplementary Figure 3. Execution time for Synthea breast cancer cohort generation across requested cohort sizes. Point labels indicate the number of generated patient rows.

Supplementary Table 4. Computational requirements for Synthea breast cancer cohort generation.

Requested Cohort Size	Generated Patient Rows	Elapsed minutes	Peak RAM (GB)	Peak RAM (MB)	CSV storage (MB)	Total storage (MB)
5	4	2.42	1.51	1548.57	15.88	16.31
10	9	4.36	1.61	1644.46	28.25	29.24
25	30	5.33	1.80	1845.95	69.49	71.07
50	57	12.94	2.05	2099.22	98.37	100.60
100	110	29.24	2.17	2217.46	179.17	182.04

Downstream post-processing performance

For downstream post-processing, the generated CSV files from the 50-patient benchmark run were used. The five documented post-processing steps were then run sequentially: clone reconstruction, clone-proportion assignment, generation of the "enhanced" observations file, passenger-mutation assignment, and complete clone-level MAF generation. Each step was measured independently with GNU "/usr/bin/time -v", and storage was measured after each step.

For the representative 50-patient generation benchmark, Synthea produced 57 patients in 12.94 minutes. The downstream post-processing benchmark for this same generated cohort required an additional 2.64 minutes in total (Supplementary Table 5 and Supplementary Table 6). Peak memory across post-processing steps was 0.89 GB. The workflow produced 65 reconstructed clone groups, 96 864 enhanced observation rows, 181 892 assigned passenger mutation rows, and 65 clone-level MAF files. Final post-processing benchmark storage was 455.66 MB, including 183.90 MB in the CSV directory, 51.61 MB of clone-level MAF files, and a 220.15 MB cohort-specific passenger mutation pool.

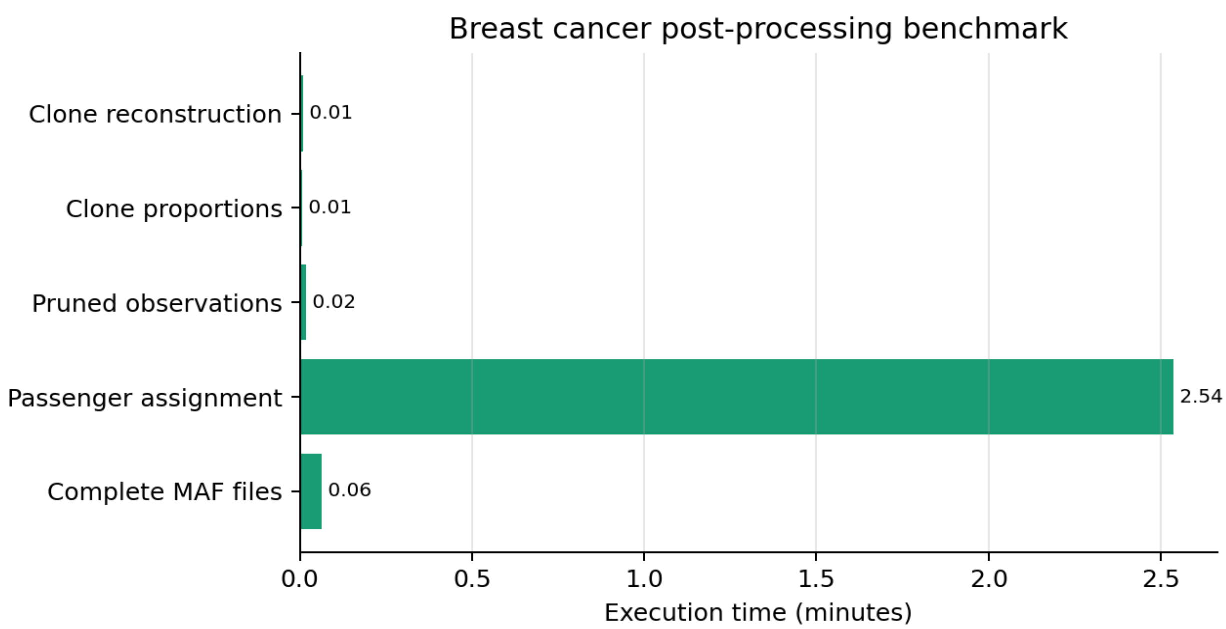
Supplementary Table 5. Computational requirements summary for downstream breast-cancer genomics post-processing of the benchmarked 50-patient cohort.

Patients	Total time (min)	Peak RAM (GB)	Assigned passenger rows	MAF Files	Final storage (MB)
57	2.64	0.89	181 892	65	455.66

Supplementary Table 6. Per-step computational requirements for downstream breast-cancer genomics post-processing of the benchmarked 50-patient cohort.

Step	Time (min)	Peak RAM (GB)	Storage after step (MB)
Clone reconstruction	0.01	0.01	98.39
Clone proportions	0.01	0.01	98.40
Enhanced observations generation	0.02	0.12	115.16
Passenger assignment	2.54	0.78	404.06
Clone-level MAF generation	0.06	0.89	455.66

The per-step runtime breakdown is shown in Supplementary Figure 4. The passenger-mutation assignment step dominated post-processing runtime, requiring 2.54 minutes of the 2.64-minute total.



Supplementary Figure 4. Runtime of downstream breast-cancer genomics post-processing steps for the representative 50-patient benchmark cohort.

Summary

Taken together, the benchmarked 50-patient workflow required approximately 15.58 minutes for Synthea generation plus genomics post-processing on the benchmark workstation. These results indicate that the framework is practical for small to moderate synthetic breast cancer cohorts on a standard workstation, while larger cohorts are expected to scale primarily with the cost of filtered Synthea generation and passenger-mutation processing. The scripts used for benchmarking can be found in the project's repository under the folder "scripts/benchmarking".

Sequencing-Level FASTQ Generation

Synthetic sequencing reads were generated with NEAT (Stephens et al. 2016) through the SyntheticCancerGenome workflow (Vazquez Garcia, n.d.) on the MareNostrum 5 GPP partition and profiled using Slurm accounting and GNU time. Computational resource usage was recorded for each synthetic tumour sequencing run. All clone-level tumour FASTQ generation jobs used the same Slurm configuration: one task, 64 requested CPUs, eight parallel NEAT chromosome-copy workers, and 16 samtools threads for consolidating the per-chromosome outputs into clone-level FASTQ files. On the MareNostrum 5 GPP partition, these jobs were allocated one node, corresponding in these runs to 128 allocated CPUs and 125 GB memory.

The computational requirements for the clone-level tumour FASTQ generation runs are summarized in Supplementary Table 7. These measurements provide practical estimates of runtime, memory use and compressed FASTQ storage requirements for generating synthetic tumour sequencing data. Tumour samples were generated at a nominal 60x total tumour depth. The simulated depth of each clone was determined from the clone fraction at each timepoint, using ceil(total tumour depth * clone fraction), which resulted in summed clone-level depths of 60x at t0 and 61x at t1 and t2.

For storage planning, compressed FASTQ output size scaled approximately with simulated sequencing depth. Across the three timepoints, the summed clone-level FASTQ output was approximately 146 GB at t0, 146 GB at t1 and 146 GB at t2. This corresponds to roughly 2.4 GB of compressed paired-end FASTQ data per simulated depth unit for this patient and read simulation configuration.

Supplementary Table 7. Representative computational requirements for clone-level synthetic tumour FASTQ generation. FASTQ sizes refer to compressed paired-end outputs. Peak RAM was measured as maximum resident set size using GNU time.

Time point	Clone	Depth	Wall-clock time	Peak RAM (GNU Time)	R1 FASTQ (.fq.gz)	R2 FASTQ (.fq.gz)	Total FASTQ Size (.fq.gz)	User CPU Time	System CPU time	Avg. CPU Use
t0	1	27x	31 h 57 min 42 s	3.3 GB	33 GB	33 GB	66 GB	594 680 s	23 789 s	537%
t0	3	33x	34 h 32 min 54 s	3.3 GB	40 GB	40 GB	80 GB	626 396 s	20 805 s	520%
t1	1	11x	20 h 15 min 43 s	3.3 GB	15 GB	15 GB	30 GB	404 542 s	20 201 s	582%
t1	2	20x	25 h 27 min 48 s	3.3 GB	23 GB	23 GB	46 GB	487 439 s	20 786 s	554%
t1	3	30x	31 h 53 min 51 s	3.3 GB	35 GB	35 GB	70 GB	586 817 s	20 881 s	529%
t2	1	14x	21 h 30 min 12 s	3.3 GB	16 GB	16 GB	32 GB	425 084 s	20 326 s	575%
t2	2	31x	33 h 13 min 38 s	3.3 GB	38 GB	38 GB	76 GB	606 367 s	21 229 s	524%
t2	4	16x	23 h 05 min 39 s	3.3 GB	19 GB	19 GB	38 GB	451 570 s	21 195 s	586%

Validation of the results

We performed two complementary validation steps on the example synthetic breast cancer dataset. First, we assessed the internal coherence of the generated clinical-genomic output files. Second, we assessed read-level recoverability of the simulated variants using a separate alignment and Mutect2-based (Benjamin et al. 2019) somatic variant-calling workflow. These validation steps were intended to evaluate consistency and technical recoverability of the generated outputs, rather than to establish clinical predictive accuracy or biological realism.

Clinical-Genomic Coherence Validation

Clinical-genomic output checks were performed using a standalone validation script ("scripts/utilities/validate_synthetic_output.py") applied to the generated output directory. The script inspected the standard Synthea clinical CSV files, the variant-aware observations, reconstructed clone groups, clone-proportion tables, assigned passenger mutation files, and clone-level MAF files. The aim was to assess whether the generated clinical breast cancer context and genomic outputs were coherent, including whether driver variants were represented in MAF files, passenger mutations were present at the expected synthetic-output scale, clone proportions were numerically valid, and the MAF files were structurally usable for downstream analyses.

Validation was performed on the representative synthetic breast cancer cohort described in the Results section, summarized in Supplementary Tables 1 and 2, and provided in the project repository. The run includes 39 synthetic patients, 48 reconstructed clones, three clone-proportion timepoints, and 48 clone-level MAF files. The MAF files contain 143 driver-labelled rows across 34 driver genes and 139 387 passenger-labelled rows. The checks confirmed that driver and passenger mutations were present in the generated MAF outputs, clone-level driver assignments were represented in the corresponding clone MAF files, clone proportions were non-negative and normalized across patient/timepoint combinations, and all clone-level MAF files contained the required columns and core variant fields.

Supplementary Table 8. Clinical-genomic validation summary for the generated synthetic breast cancer dataset.

Validation domain	Result	Key evidence	Interpretation
Breast cancer clinical context	PASS	39/39 synthetic patients had a malignant breast cancer diagnosis.	The genomic MAF outputs were generated for patients with the

			intended breast cancer clinical context.
Driver variant representation	PASS	The clone-level MAF files contained 143 driver-labelled rows across the 34 pre-defined driver genes. Common driver genes included PIK3CA, GATA3, MAP3K1, ESR1, TP53, CCND1, BRCA2, ERBB2, ZNF703, and CHEK2.	Driver variants are present in the MAF files and cover clinically relevant breast cancer genes.
Clone-to-MAF driver consistency	PASS	All 143 clone-level driver gene assignments were present in the corresponding clone-level MAF files.	Driver genes assigned to each reconstructed clone were carried through into that clone's mutation file.
Clinical biomarker and genomic observation context	PASS	The "enhanced" observations file contained 1 170 mutation-related rows, out of which 234 were genomic DNA change rows, 234 were clonal prevalence rows, and 234 were clonal role rows (either founding or later).	The clinical observation output contains breast cancer biomarker and mutation context that can be interpreted alongside the clone and MAF outputs.
Passenger mutation representation	PASS	The clone-level MAF files contained 139 387 passenger-labelled rows. Passenger burden per patient ranged from 1692 to 54 748 variants, with a median of 5008. Passenger burden per clone ranged from 464 to 27	Passenger mutations are present at substantial levels in the generated MAF files. The counts reflect transferred ICGC-derived passenger mutation sets.

		374 variants, with a median of 1607.	
Clone abundance trajectories	PASS	Clone proportions were non-negative, non-zero, and all patient/timepoint clone proportions sum to 100%.	The clone abundance trajectories are numerically coherent across sequencing timepoints.
MAF structural validity	PASS	All 48 clone-level MAF files contained required columns, and none of the MAF rows had missing core variant fields.	The MAF files are structurally usable for downstream mutation-level analysis.
Variant burden distribution	PASS	Across 48 clone MAF files, total variants per clone ranged from 466 to 27 379. Across the 18 patients with MAF output, total variants per patient ranged from 1696 to 54 756.	Variant burdens are heterogeneous across clones and patients, as expected after assigning patient-specific passenger mutation sets and clone-level driver variants.

FASTQ-Level Validation by Somatic Variant Calling

To validate that simulated somatic mutations were present in the generated sequencing reads, each merged tumour FASTQ pair was processed through the same independent alignment and somatic variant-calling workflow. Validation was performed using one shared 30x matched-normal sample and one merged tumour sample per timepoint at a nominal target depth of 60x. Reads were aligned to a standard hg38 reference using BWA-MEM (Li, 2013), and the resulting BAM files were sorted and indexed with samtools (Danecek et al. 2021). BAM integrity was assessed using samtools quickcheck, and alignment summaries were generated using samtools flagstat.

For each timepoint, a truth set was constructed from the expected hg38 somatic variants of the active clones present in that tumour sample. The corresponding hg38 VCFs were combined, normalized and de-duplicated. Mutect2 was run in matched tumour-normal mode using the aligned tumour and normal BAM files. To make validation focused and computationally efficient, variant calling was restricted to padded intervals around the expected truth variants. FilterMutectCalls (Van der Auwera and O'Connor 2020) was applied to obtain filtered somatic calls, and PASS calls were normalized and compared against the timepoint-specific truth set using bcftools (Danecek et al. 2021).

This validation was intended to test whether expected simulated variants were recoverable from the generated FASTQ files after alignment and variant calling. Recovery was first assessed at the level of filtered PASS Mutect2 calls. Because variants not recovered as PASS calls may still be present in the reads but removed by variant-caller filtering, a targeted read-support analysis was also performed at all expected truth variant sites. For each truth variant, the aligned tumour and matched normal BAM files were inspected directly, and reference- and alternate-supporting reads were counted. This allowed variants missed by Mutect2 to be separated into those with direct tumour alternate-allele support and those without detectable tumour alternate-allele support in the aligned reads.

Supplementary Table 9. FASTQ-level validation of the simulated timepoint 0 tumour sample using matched tumour-normal somatic variant calling and targeted read-support analysis at expected truth variant sites. The results correspond to 91.44% PASS recovery. Overall, 864/935 expected variants (92.41%) had direct tumour ALT support. Among the 80 variants missed by Mutect2, 9 (11.25%) were nevertheless present in aligned tumour reads, while 71 had no detected ALT support.

Validation metric	Result
Normal BAM Size	68 GB
Tumour t0 BAM Size	138 GB
Normal mapped reads	700 253 745
Tumour mapped reads	1 447 177 804
Tumour read pairs	723 578 313
BAM integrity check	Passed
Expected t0 truth variants	935

PASS Mutect2 calls	889
Truth variants recovered as PASS	855
Truth variants not recovered as PASS	80
PASS calls outside truth set	34
Truth variants with tumour ALT read support	864
Mutect2-missed truth variants with tumour ALT support	9
Mutect2-missed truth variants without tumour ALT support	71

Supplementary Table 10. FASTQ-level validation of the simulated timepoint 1 tumour sample using matched tumour-normal somatic variant calling and targeted read-support analysis at expected truth variant sites. The results correspond to 91.37% PASS recovery. Overall, 1298/1402 expected variants (92.58%) had direct tumour ALT support. Among the 121 variants missed by Mutect2, 17 (14.05%) were nevertheless present in aligned tumour reads, while 104 had no detected ALT support.

Validation metric	Result
Normal BAM Size	68 GB
Tumour t1 BAM Size	138 GB
Normal mapped reads	700 253 745
Tumour mapped reads	1 447 373 043
Tumour read pairs	723 687 908
BAM integrity check	Passed
Expected t1 truth variants	1402
PASS Mutect2 calls	1328
Truth variants recovered as PASS	1281

Truth variants not recovered as PASS	121
PASS calls outside truth set	47
Truth variants with tumour ALT read support	1298
Mutect2-missed truth variants with tumour ALT support	17
Mutect2-missed truth variants without tumour ALT support	104

Supplementary Table 11. FASTQ-level validation of the simulated timepoint 2 tumour sample using matched tumour-normal somatic variant calling and targeted read-support analysis at expected truth variant sites. The results correspond to 90.71% PASS recovery. Overall, 1305/1400 expected variants (93.21%) had direct tumour ALT support. Among the 130 variants missed by Mutect2, 35 (26.92%) were nevertheless present in aligned tumour reads, while 95 had no detected ALT support.

Validation metric	Result
Normal BAM Size	68 GB
Tumour t2 BAM Size	138 GB
Normal mapped reads	700 253 745
Tumour mapped reads	1 447 372 981
Tumour read pairs	723 687 857
BAM integrity check	Passed
Expected t2 truth variants	1 400
PASS Mutect2 calls	1 312
Truth variants recovered as PASS	1 270
Truth variants not recovered as PASS	130
PASS calls outside truth set	42

Truth variants with tumour ALT read support	1 305
Mutect2-missed truth variants with tumour ALT support	35
Mutect2-missed truth variants without tumour ALT support	95

Supplementary Methods: Additional Implementation Details

The Methods section in the main text describes the complete workflow used to generate the linked synthetic clinical-genomic breast cancer dataset. This section provides only additional implementation details.

Driver-gene sampling and treatment-related assignment details

For driver-gene sampling, the PCAWG/ICGC panorama of driver-mutation file (ICGC ARGO, n.d.; Zhang et al. 2019; The ICGC/TCGA Pan-Cancer Analysis of Whole Genomes Consortium et al. 2020) was treated as an event list, where each row represented a driver event in a tumour sample. Gene-specific sampling probabilities for the non-treatment-specific driver gene set were calculated from the number of unique breast cancer samples in which each gene appeared at least once. The PCAWG/ICGC-derived gene frequencies were precomputed into the weighted driver-gene distributions used by the Synthea driver-assignment module, rather than recalculated during every run. The configured driver-count distribution allowed up to 16 assigned driver genes per tumour.

The two-draw structure used for treatment-related driver assignment was implemented to increase patient-level diversity and avoid assigning the same complete treatment-related gene set to every eligible patient. Genes supported by sensitivity or resistance evidence for more therapies within the same treatment group were prioritized in the first draw, while additional compatible genes could be assigned in the second draw when allowed by the treatment-related driver budget.

Clone reconstruction and clonal proportion details

Clone reconstruction followed a simplified tumour evolution model in which the founding clone was intended to represent early truncal events. Genes with informative longitudinal behaviour were grouped into later clones when they shared similar clonal-trend patterns across available tumour genomic profiling timepoints (e.g. all decreasing genes were grouped into one clone). Genes with non-informative or unknown clonal trends were merged heuristically into existing founding or later clones, keeping the maximum allowed number of genes in a clone up to five. Unusually large clone groups were split to avoid unrealistic clone structures.

During clonal proportion assignment, additional rule-based adjustments were applied after the initial trend-based abundance values were generated. These adjustments limited unrealistic midpoint and final abundances for decreasing clones and used the founding clone as a buffer when proportions had to be redistributed to preserve a total of 100%. Genes assigned to the same clone shared the same longitudinal abundance pattern and therefore the same estimated clone proportion at each timepoint. Clonal proportions were rounded to two decimal places and normalized so that clone percentages summed to exactly 100% at each tumour timepoint.

Variant-mapping details

CIViC-derived (Griffith et al. 2017) therapy-specific variants were matched to ICGC MAF rows using exact allele information, including chromosome, start position, end position, reference allele and alternate allele. CIViC molecular profiles that could not be matched to an exact MAF row were excluded from automatic assignment.

When multiple compatible therapy-specific variants were available for the same patient-gene-treatment context, one was selected randomly. In contrast, fallback selection from the general driver variant library was deterministic for each patient-gene combination.

Sources of stochasticity and deterministic post-processing

The pipeline contains stochastic steps, so different executions can produce different synthetic cohorts. Variability comes from the underlying Synthea (Walonoski et al. 2018) patient simulation, the custom driver-assignment module, sampling from treatment-related and non-treatment-specific driver gene pools, and random selection among eligible CIViC-compatible therapy-specific variants. As a result, cohort composition, assigned driver genes, selected variants and reconstructed clonal structures may differ between runs.

Some deterministic steps were used where reproducibility was important. General driver-variant fallback selection was seeded from the patient-gene combination, so the same patient and gene receive the same fallback variant across repeated post-processing runs. Passenger mutation assignment was also deterministic at the patient level: each synthetic patient was linked to one source breast cancer sample from the passenger-only ICGC pool, and that sample's passenger mutations were transferred to the patient. During clone-level MAF generation, passenger mutations were split across clones using a patient-specific deterministic seed, with each passenger mutation assigned to only one clone file.

Sequencing-level implementation details

For sequencing-level generation, a patient-specific manifest was created to define the simulation inputs. The manifest included the patient identifier, sex, target normal and tumour sequencing depths, active clone identifiers, clone fractions, clone type, parent clone information and paths to clone-specific MAF files. Clone fractions were normalized within each selected timepoint, and clone-specific tumour depths were calculated from the total tumour depth and rounded upward to the nearest integer. This manifest ensured that the matched normal sample and all tumour clones used the same patient-specific reference and germline background.

The germline generation workflow sampled variants from the 1000 Genomes Phase 3 VCF (The 1000 Genomes Project Consortium et al. 2015) using population allele frequencies (the EUR_AF field). It generated paternal and maternal haploid contributions, lifted selected positions from hg19 to hg38, validated them against the hg38 reference and combined them into a diploid germline mutation list.

Additional processing was applied to make the reference and germline sex-aware. For male patients, autosomes were represented as diploid, while chromosomes X, Y and mitochondrial DNA were represented as haploid. For female patients, autosomes and chromosome X were represented as diploid, chromosome Y was removed and mitochondrial DNA was retained as haploid. The germline mutation file was filtered so that all retained variants matched contigs present in the corresponding sex-aware reference.

Somatic variants from clone-specific MAF files were converted into the mutation format required by SyntheticCancerGenome. MAF records were converted to VCF-like records, normalized against hg19, lifted to hg38, normalized again against hg38, split when multi-allelic and filtered to remove failed liftovers, unsupported contigs and reference/allele mismatches. Cumulative mutation files were then generated for each clone by following the clone lineage and combining ancestor and clone-specific mutation lists while removing duplicate mutation records. The FASTQ-generation wrappers wrote metadata files recording the reference, germline, somatic mutation file, simulated depth, Rbbt job path and output FASTQ paths used for each normal or clone-level tumour simulation.

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